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Broad-Role Codebook Transfer

Cross-folio fixed-training tournament over 5,040 role mappings

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Framework by Christine Preedy | Independent testing, reconstruction, and report by Juan Gabriel Molina.

Abstract

This report documents a retrospective fixed-label reconstruction of the Preedy broad-role encoding-transfer test. An order-2 categorical count model is trained on 13,461 noninitial glyph targets across 39 source folios and evaluated on 3,262 exported scoring positions from ten nonoverlapping holdout folios, f22r-f26v. Under the fixed training encoding, the unchanged seven-role mapping ranks first among all 5,040 permutations, with no tied rival, held-out log loss 1.164825 nats per target and 52.0846% exact next-role accuracy. The next-best mapping swaps PLANET and DOSE and has loss 1.460386. The identity mapping also improves on first-order and unigram baselines. A necessary symmetry control changes each mapping consistently in training and testing, refits the same model, and makes all 5,040 variants tie exactly. The result therefore supports stable cross-folio encoding transfer under this fixed representation; it does not independently identify the literal English meanings of the role labels.

Research question and scope

Does the role encoding learned on the 39-folio source sample transfer unchanged to the held-out f22r-f26v scoring set better than every test-time permutation of the seven principal broad roles? The experiment is a fixed-training prediction tournament, not a semantic translation contest. It was reconstructed after prior results were known because the original historical runner was unavailable; no prospective status is claimed for this reconstruction.

Keywords: Voynich Manuscript; Preedy framework; held-out prediction; rival codebook; cross-folio transfer; reproducibility

1. Data, split and representation

The training table contains the primary-eligible occurrence records from the recovered source export. Blocks are identified by block ID and source spreadsheet row; glyph order is the stored glyph index. A target is scored for every noninitial glyph. The holdout table contains the previously exported two-role contexts and next raw glyph for ten folios f22r through f26v. Normalized folio identifiers are disjoint between training and testing.

Property	Frozen specification
Training folios	39
Training noninitial targets	13,461
Holdout folios	10 (f22r-f26v)
Holdout scoring positions	3,262
Primary permuted roles	PLANET, GATE, PROJECT, RECEIVE, STRING, DOSE, OTHER
Other roles	HORIZON, SOLAR and SPECIAL remain fixed
Smoothing	alpha = 0.2

1.1 Why the split matters

The rank is driven by prediction on folios not present in the training sample. This blocks direct reuse of the same folio rows for both parameter fitting and test loss. It does not make the full scientific study prospective: the representation itself was developed before this reconstruction, and the historical holdout was not virgin to all prior project work.

1.2 Broad-role target

The model predicts the next broad role from the two preceding broad roles, with BOS used where the second previous role does not exist. The test asks whether the trained encoding remains internally aligned across folios. Because the labels are categorical symbols, their English names require a separate semantic test.

2. Predictive model and tournament

For each pair of previous roles, the model stores counts of the next role. Additive smoothing is applied to every outcome. If a two-role context is absent from training, the prediction backs off to the corresponding one-role distribution; if that one-role context is also absent, it backs off to the unigram distribution.

P(next = k | r[t-2], r[t-1]) = (count + alpha) / (context total + alpha K), with alpha = 0.2.

The model is intentionally explicit and low-dimensional. No position bin, block-length bin, folio identity, visual feature, anatomy label or language model is used. No hyperparameter search was conducted for this release.

2.1 Rival construction

All 7! = 5,040 permutations of the seven principal broad roles are applied to the held-out contexts and held-out target labels while the fitted training model remains fixed. The identity mapping is the original encoding. Lower mean negative log probability is better. Exact-role accuracy is reported descriptively but does not define the ranking.

2.2 Symmetry control

A categorical model must not be treated as semantic evidence merely because the identity labels rank first under a fixed encoding. The decisive control therefore applies each permutation consistently to training and test data, refits, and rescales the labels everywhere. Under a label-equivariant count model, those versions should be mathematically equivalent. They are: the maximum loss difference across all consistently relabeled variants is exactly 0.0 in the stored numerical output.

This control changes the interpretation of the tournament. A fixed-training win is evidence that the held-out corpus uses the same learned encoding. It is not evidence that the names PLANET, GATE or DOSE have been externally identified.

3. Primary results

Metric	Result
Identity rank	1 / 5,040
Tied rivals	0
Held-out loss	1.1648254357 nats/target
Exact next-role accuracy	52.0846%
First-order baseline loss	1.2837544693
Unigram baseline loss	1.9458876050
Next-best rival	PLANET <-> DOSE swap
Next-best loss	1.4603860843
Consistent relabel max difference	0.0

The unchanged encoding is separated from the nearest test-time rival by 0.295561 nats per target. Relative to the first-order baseline, the order-2 identity loss improves by 0.118929 nats per target; relative to the unigram baseline, it improves by 0.781062. These are descriptive log-loss differences and should not be converted into probabilities that the key is correct.

3.1 Five best tournament entries

Rank	Mapping summary	Loss	Accuracy
1	identity	1.164825	52.08%
2	PLANET->DOSE; GATE->GATE; PROJECT->PROJECT; RECEIVE->RECEIVE; STRING->STRING; DOSE->PLANET; OTHER->OTHER	1.460386	45.03%
3	PLANET->PLANET; GATE->GATE; PROJECT->PROJECT; RECEIVE->RECEIVE; STRING->OTHER; DOSE->DOSE; OTHER->STRING	1.574193	44.39%
4	PLANET->OTHER; GATE->GATE; PROJECT->PROJECT; RECEIVE->RECEIVE; STRING->STRING; DOSE->DOSE; OTHER->PLANET	1.587617	45.89%
5	PLANET->PLANET; GATE->GATE; PROJECT->PROJECT; RECEIVE->RECEIVE; STRING->STRING; DOSE->OTHER; OTHER->DOSE	1.593658	50.98%

Table 1. The complete 5,040-row candidate table is included in the package. The top-five table is descriptive; only the full exhaustive tournament determines the rank.

4. Folio-level transfer

Folio	Rows	Order-2 loss	Unigram loss
f22r	411	1.050679	1.913579
f22v	310	0.972342	1.938605
f23r	365	1.123949	1.938787
f23v	325	1.211974	1.995771
f24r	352	1.268694	2.031122
f24v	326	1.220708	1.988257
f25r	211	1.147745	1.917474
f25v	234	1.104716	1.943925
f26r	353	1.225948	1.888450
f26v	375	1.291476	1.905438

Table 2. Every holdout folio has lower order-2 loss than its corresponding unigram loss. The aggregate tournament rank is still the primary endpoint; folio rows are supplied to show that the total result is not produced by a single folio.

4.1 What the result supports

The held-out f22r-f26v rows are best predicted when their seven principal role labels retain the encoding learned from the source folios. Within this reconstructed model, no alternate test-time reassignment of those seven labels performs as well.

4.2 What the result does not support

The exhaustive rank is not a p-value. The 5,040 candidates are deterministic relabelings, not random draws from a population of semantic theories. The symmetry control proves that an arbitrary global renaming of the categorical states cannot be distinguished when training and testing are renamed together. Accordingly, the report does not claim that this experiment alone establishes planets, gates, strings, dosing or any literal English referent.

5. Limitations and experiment boundary

5.1 Retrospective reconstruction

The original historical model/runner was not recovered. This release therefore uses an explicit order-2 count model frozen before the September 11 reconstruction run. Prior project results were already known. The result is a new reproducible reconstruction, not a time-stamped replay of the historical experiment.

5.2 Representation dependence

The outcome is conditional on the supplied source encoding, the seven selected role categories, the exported holdout rows and this particular predictive model. Other representations or models could produce different rival orderings. The experiment does not compare the Preedy partition against every known Voynich slot model or against every possible alphabet partition.

5.3 No independent semantics

Stable encoding transfer is necessary for a reusable codebook, but it is not sufficient for decipherment. Literal meanings require independent anchors, external prediction or cross-modal evidence that breaks categorical label symmetry.

5.4 No multiplication with other tests

This result shares source assumptions with the structural-null and other Preedy analyses. Its rank should not be multiplied with their p-values or treated as an independent probability of decipherment. Each experiment answers a narrower question.

5.5 Reproducibility versus independent replication

Executing the packaged runner again reproduces this reconstructed fixed-label tournament. That is a reproducibility check on the same frozen representation and data; it is not a statistically or historically independent replication of the original experiment.

6. Reproducibility, provenance and references

The accompanying package contains the two source CSVs, a standalone runner, the full 5,040-candidate table, folio-level losses, result JSON, protocol, requirements file and SHA-256 manifest.

```
python -m pip install -r requirements.txt
python code/run_codebook_transfer.py
python verify_manifest.py
```

The runner regenerates the tournament and writes `results/all_candidates.csv`, `results/per_folio.csv` and `results/result.json`. The report headline was checked against the regenerated result before packaging.

Package data fingerprints:
source_occurrences.csv: e7f2dc706365d360edfa9bbf97ca0702db5377eb44ace718515bf326f7470bbd
holdout_scoring_data.csv: 17783c60ee3898aaf0c33b0b24547078009e1a21fea22a3be667100f4b4e96d8

References and source records

[1] Preedy, C. and Molina, J. G. (2026). Broad-Role Codebook Transfer Handoff, version 2026-09-11. README.md and PROTOCOL.md.

[2] Same package. data/source_occurrences.csv; data/holdout_scoring_data.csv; code/run_codebook_transfer.py.

[3] Same package. results/result.json; results/all_candidates.csv; results/per_folio.csv.

Contributions: Christine Preedy originated the framework and proposed interpretations. Juan Gabriel Molina contributed the testing architecture, computational analysis, reproducibility packaging and report.